



11/10/24

CRATER

Caltech Air and
Outer Space



We are



CRATER



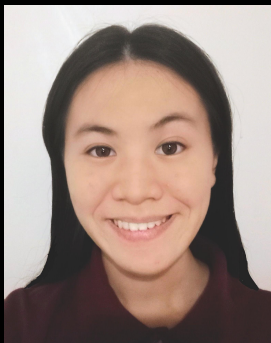
Rover at CIRC



Leads



Randy Ngo



Ruth Berkun



Jihyun Jeon



Kevin Gauld



Bella Kwaterski

URC & CIRC

What Is URC?



- ▶ The **University Rover Challenge** has been held yearly by the Mars Society in the Utah desert since 2007
- ▶ 36 teams are selected worldwide to compete at the Mars Desert Research Station, Hanksville, Utah
- ▶ The rover must be capable of accomplishing several **remote missions** posing different challenges in autonomy, locomotion, actuation, and more

URC - Delivery

- Pick up and carry 5kg payload
- Find equipment in a large search area
- Find objects with poor radio reception

URC - Equipment Servicing

- Pick up test tube and insert into cache
- Autonomous typing
- Open a latch and drawer
- Pick up and insert a USB stick
- Operate a joystick
- Push buttons, flip switch, turn knobs

URC - Autonomous Navigation

- 2 Locations with GNSS coordinates
- 3 Locations with approximate GNSS, AR tags
- 3 Locations with approximate GNSS, Object on ground

URC - Science

- Select sites of biological interest from data collected by rover
- Capture various photos
- Life detection capability
- Capability to measure soil humidity and temperature at least 10 cm below surface
- Collect subsurface sample, at least 5g

What Is CIRC?



*Similar missions, different science task

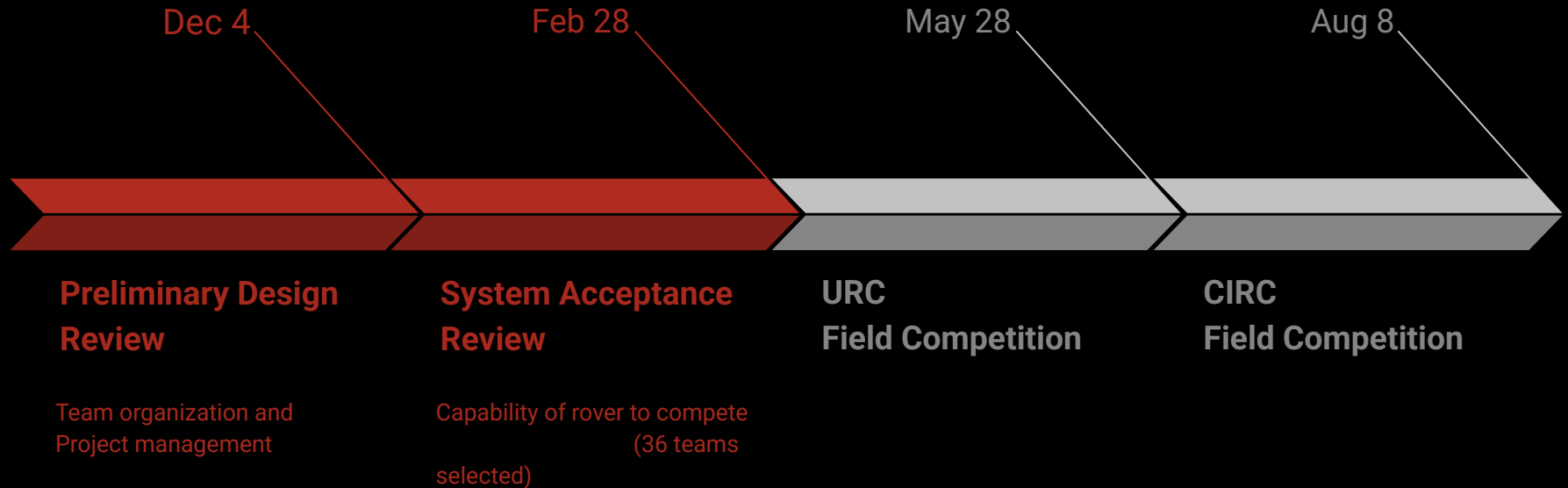
- ▶ The **Canadian International Rover Challenge** has been held twice a year by the Canadian Space Technology Advancement Group (CSTAG) since 2017
- ▶ Teams from around the world to compete at Drumheller, Alberta, Canada
- ▶ The rover must be capable of accomplishing several **remote missions** posing different challenges in autonomy, locomotion, actuation, and more

Club Organization

Structure

- ▶ Project groups of 2-3 people, mix of experience
- ▶ Subteam meetings next week for project selection
- ▶ Update meetings for related projects
(ex: manipulator, science, electrical, comms)

Timeline



Worksessions every Sunday 2-4pm, Project group meetings as needed

Projects

Mechanical

- Designing Gripper
- Designing Teststand
- Designing Custom Gearbox
- Designing Science end effector *Sc
- Redesigning for new loads *A
- Motor selection for direct drive transmission
- Comms antenna stand *E

Software

- Lower level motor code
- Autonomous Exploration
- Arm Kinematics and control *A
- Path Planning
- Vision *Sc

Electrical/Communications

- comms lines for video and rover control *S
- Upgrade BMS/battery system
- Automatic tracking antenna *M

Science

- Astrobiology!
- Testing for signs of life
- Sampling terrain *M,A
- Panoramic photos *S

How to Get Involved

Upcoming Subteam Meetings

Nov 17 (Sun), Keck 111

Talk more in depth and choose projects

- ▶ Mechanical - 2:00 ~ 2:30 pm
- ▶ Software - 2:30 ~ 3:00 pm
- ▶ Electrical - 3:00 ~ 3:30 pm
- ▶ Science - 3:30 ~ 4:00 pm

Questions?



<https://bit.ly/crater-2024>



<https://discord.gg/d9KwesEA>

